Research Article

Long-run Impacts of Geopolitical Risk and Economic Policy Uncertainty on Foreign Direct Investments in Türkiye

Faruk MİKE[1*], İbrahim Ethem AKYILDIZ[2], Oktay KIZILKAYA[3]

[1] Assoc. Prof. Dr., Osmaniye Korkut Ata University, Bahçe Vocational School, farukmike@osmaniye.edu.tr, ORCID: 0000-0002-9194-1679
[2] Assist. Prof. Dr., Bursa Uludağ University, FEAS, Department of Economics, ethemakyildiz@uludag.edu.tr, ORCID: 0000-0001-6850-115X
[3] Prof. Dr., Malatya Turgut Özal University, Department of International Trade and Finance, oktay.kizilkaya@ozal.edu.tr, ORCID: 0000-0002-3412-5616
[*] Corresponding Author: farukmike@osmaniye.edu.tr

Abstract: This study examines the determinants of foreign direct investment (FDI) in Türkiye over the period 1985–2024, with a particular focus on economic policy uncertainty and geopolitical risk. In addition to these key risk factors, the analysis also incorporates economic growth and institutional quality as control variables within a long-run empirical framework. To account for possible structural changes in the Turkish economy, the study employs advanced time-series econometric techniques that consider fractional frequency values. The empirical findings reveal that both economic policy uncertainty and geopolitical risk have statistically significant and negative effects on FDI inflows. These results indicate that increases in uncertainty and geopolitical tensions discourage foreign investors by raising perceived risk and weakening investment confidence. In contrast, economic growth and institutional quality are found to exert positive effects on FDI, although their influence is secondary compared to risk-related variables. Overall, the results suggest that FDI inflows to Türkiye are primarily driven by risk and uncertainty conditions rather than purely economic fundamentals. The findings highlight the importance of macroeconomic stability, policy predictability, and geopolitical risk mitigation in attracting foreign capital. From a policy perspective, reducing economic policy uncertainty and strengthening geopolitical stability emerge as key priorities for improving Türkiye's investment climate.

Keywords: Foreign Direct Investment, Economic Policy Uncertainty, Geopolitical Risk, Institutional Quality, Türkiye
Jel Codes: F21, D80, F51, C32

Cite: Mike, F.; Akyıldız, İ., E.; Kızılkaya, O. (2026). Long-run Impacts of Geopolitical Risk and Economic Policy Uncertainty on Foreign Direct Investments in Türkiye, *Politik Ekonomik Kuram*, *10*(2), 875-891. https://doi.org/10.30586/pek.1942446

Submitted: 02.05.2026
Accepted: 25.06.2026

Copyright: © 2026. (CC BY) (https://creativecommons.org/licenses/by/4.0/).

Jeopolitik Risk ve Ekonomik Politika Belirsizliğinin Türkiye'de Doğrudan Yabancı Yatırımlar Üzerindeki Uzun Dönemli Etkileri

Öz: Bu çalışma, 1985–2024 dönemi için Türkiye'de doğrudan yabancı yatırımların belirleyicilerini ekonomik politika belirsizliği ve jeopolitik risk odağında incelemektedir. Söz konusu temel risk unsurlarına ek olarak analizde, uzun dönemli ampirik çerçeve kapsamında iktisadi büyüme ve kurumsal kalite kontrol değişkenleri olarak modele dâhil edilmiştir. Türkiye ekonomisinde ortaya çıkabilecek yapısal değişimleri dikkate almak amacıyla çalışmada, kesirli frekans değerlerini göz önünde bulunduran gelişmiş zaman serisi ekonometrisi yöntemleri kullanılmıştır. Ampirik bulgular, ekonomik politika belirsizliği ve jeopolitik riskin doğrudan yabancı yatırım girişleri üzerinde istatistiksel olarak anlamlı ve negatif etkiler yarattığını göstermektedir. Bu sonuçlar, belirsizlikteki artışların ve jeopolitik gerilimlerin, algılanan riski yükselterek ve yatırımcı güvenini zayıflatarak yabancı yatırımcıları caydırdığını ortaya koymaktadır. Buna karşılık, iktisadi büyüme ve kurumsal kalitenin doğrudan yabancı yatırımlar üzerinde pozitif etkiler oluşturduğu, ancak bu etkilerin risk ve belirsizlik değişkenlerine kıyasla ikincil düzeyde kaldığı görülmektedir. Genel olarak bulgular, Türkiye'ye yönelik doğrudan yabancı yatırım girişlerinin yalnızca ekonomik temellerden ziyade, büyük ölçüde risk ve belirsizlik koşulları tarafından şekillendiğini göstermektedir. Çalışmanın sonuçları, yabancı sermayenin çekilmesinde makroekonomik istikrarın, politika öngörülebilirliğinin ve jeopolitik risklerin azaltılmasının önemini

vurgulamaktadır. Politika açısından değerlendirildiğinde, ekonomik politika belirsizliğinin azaltılması ve jeopolitik istikrarın güçlendirilmesi, Türkiye'nin yatırım ortamını iyileştirmeye yönelik temel öncelikler olarak öne çıkmaktadır.

Anahtar Kelimeler: Doğrudan Yabancı Yatırımlar, Ekonomik Politika Belirsizliği, Jeopolitik Risk, Kurumsal Kalite, Türkiye
Jel Kodları: F21, D80, F51, C32

1. Introduction

The world has faced a series of significant social, economic, and political disruptions over the past quarter century. Tensions among major actors in the international economic and political system constitute one of the most critical dimensions of these disruptions. Trade tensions between China and the United States, the Russia–Ukraine war, India–Pakistan tensions, and the Iran–Israel conflict are among the most prominent geopolitical examples of this period. In addition, developments such as the devastation in Gaza, the United Kingdom's withdrawal from the European Union, the COVID-19 shock, and the 2008 global financial crisis have also had profound effects on the global system. These processes have fuelled nationalism and populism, increased uncertainty in the global political economy, weakened the capacity for international response to the COVID-19 pandemic, and made the fragile structure of the international relations system more visible (Yu and Wang, 2023). As a result, these developments have led to significant fluctuations in the world economy and an increase in recessionary pressures.

UNCTAD's World Investment Report 2025 explains these challenges facing the global economic system through three main dilemmas: rising trade tensions, policy uncertainty, and geopolitical fragmentation. According to the report, global foreign direct investment (FDI) declined by 11% in 2024, falling to approximately 1.5 trillion dollars. This decline is particularly driven by the slowdown in infrastructure investments and the contraction in industrial investments. It is also emphasized that if current global challenges persist, investment environments may deteriorate further, and developing countries, which are most dependent on capital inflows, are expected to be negatively affected (UNCTAD, 2025).

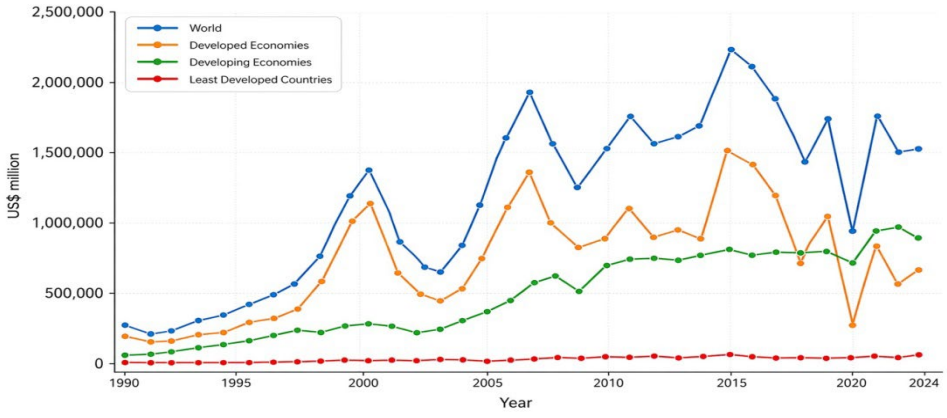


Figure 1. Trends in Global Foreign Direct Investment Inflows, 1990-2024 (Source: UNCTAD, www.unctad.org/fdistatistics)

Figure 1 presents the historical global outlook of FDI flows. It shows that global FDI inflows between 1990 and 2024 have followed a volatile pattern and are highly sensitive to changes in global economic conditions. Although FDI inflows have generally exhibited

an upward long-term trend, significant declines have been observed during crisis periods, particularly around the 2008 financial turmoil and the pandemic episode in 2020. While developed economies maintain their dominant position in global FDI flows, they also display fluctuations over time. In contrast, the share of developing countries in global investment flows has gradually increased. Least developed countries, however, continue to account for a very limited share throughout the period. Despite this progress, it is noteworthy that total global investment volume remains below its peak level reached in 2015. Overall, this pattern indicates that global capital flows have become more widely distributed across countries, while at the same time remaining vulnerable to global shocks.

Türkiye, due to its geopolitical position and its role in the global economy, often finds itself at the center of these global challenges. As is well known, its geographical location at the crossroads of three continents places it in a strategic position between the politically unstable Middle Eastern countries and democratic European economies (Yılancı et al., 2026). At the same time, with a nominal gross domestic product of approximately 1.32 trillion dollars as of 2024, Türkiye ranks as the 17th largest economy in the world and constitutes an important actor in the global economy (World Bank, 2026). While these geopolitical and economic characteristics create significant opportunities for the Turkish economy, they also increase its vulnerability to external shocks and various fragilities.

Türkiye is arguably one of the countries most exposed to geopolitical risk and economic uncertainty in the world. In terms of geopolitical risks, prolonged political instability and violent conflicts in neighboring regions have been key determinants. The 2003 US–Iraq war, the 2011 Syrian civil unrest, the 2022 Russia–Ukraine war, and the recent Iran–Israel conflict are among the most significant examples. These conflicts have generated multidimensional risks, particularly in terms of migration flows, border security, regional stability, and national security policies (Kızılkaya et al., 2026). In terms of economic uncertainty, both global crises and recent domestic economic and political policies have played important roles. At times, inconsistencies between monetary and fiscal policy have contributed to cycles of inflation, exchange rate volatility, and high interest rates, thereby significantly increasing uncertainty in the economy. Overall, these indicators suggest that macroeconomic stability in Türkiye is strongly constrained by geopolitical risks and domestic policy uncertainty.

Beyond risk and uncertainty factors, FDI inflows are also shaped by a broader set of economic and institutional determinants. Market size and growth performance, trade openness, macroeconomic stability, labour market conditions, infrastructure quality, exchange rate stability, tax incentives, financial development, and the overall institutional environment are commonly regarded as important factors influencing foreign investors' location decisions. In particular, larger and faster-growing markets may attract efficiency- and market-seeking investments, while stronger institutions, regulatory quality, and policy credibility reduce transaction costs and improve the predictability of the investment environment. However, the relevance of these determinants may weaken when investors face high levels of policy uncertainty and geopolitical risk. Therefore, while this study acknowledges the broader determinants of FDI, it specifically focuses on economic policy uncertainty and geopolitical risk as two critical risk-related channels shaping FDI inflows to Türkiye.

Motivated by this background, the present study examines how two major sources of investment risk—geopolitical tensions and uncertainty surrounding economic policy—shape FDI inflows to Türkiye. As a developing country with significant economic potential, Türkiye remains heavily dependent on foreign investment, particularly in infrastructure and technology sectors. Changes in these risk factors can have substantial effects on FDI inflows and therefore play a decisive role in the country's growth and development process. This highlights the importance of the findings for policy implications.

This study makes three main contributions. First, the study advances the existing literature by assessing, within a unified empirical framework, how geopolitical risk and economic policy uncertainty shape FDI inflows to Türkiye. While existing studies generally analyze these risk factors separately, this study provides a more comprehensive framework by considering their simultaneous effects. Second, it strengthens the empirical evidence on the risk–investment nexus by focusing on Türkiye during a period characterized by intense global and regional shocks over the past quarter century. In this way, it contributes to the literature on the sensitivity of capital flows to external shocks. Third, the results yield policy-relevant evidence by clarifying the extent to which geopolitical risk and policy-related uncertainty constrain FDI inflows, thereby underscoring the need for a more stable macroeconomic environment, a more predictable policy framework, and a more credible investment climate.

The structure of the study is as follows: The section 2 presents the theoretical relationship between geopolitical risk, economic policy uncertainty, and FDI. The remainder of the study is organized as follows. Section 3 synthesizes the relevant empirical literature, while Section 4 describes the data and variables used in the analysis. Section 5 outlines the econometric strategy, followed by the presentation of the empirical results in Section 6. Section 7 discusses the main findings in light of the existing literature, and the final section concludes by deriving policy implications from the empirical evidence.

2. Theoretical Framework

FDI plays a critical role, particularly for developing countries that face chronic shortages of technology and capital. These investments contribute directly to economic growth and development by improving labor markets, increasing employment, and enhancing productivity. For this reason, improving the investment climate and increasing FDI inflows constitute key policy priorities for many developing economies (Quazi, 2007). However, economic, political, and social changes at the global level significantly affect the direction and volume of these capital flows. This situation leads to more pronounced negative effects of external shocks, especially for economies dependent on foreign capital.

Among these global shocks, geopolitical risks and economic policy uncertainty have recently emerged as two of the most influential factors affecting national economies and capital movements. This mechanism can also be interpreted through the real options investment framework. According to Dixit and Pindyck (1994), when investment decisions are partly irreversible and future conditions are uncertain, firms may prefer to postpone investment because uncertainty increases the option value of waiting. A common feature of both is their ability to generate risk and uncertainty conditions that alter investor behavior. For instance, geopolitical risks can discourage FDI by increasing investment costs, reducing predictability, and raising institutional risks. They may also lead multinational corporations to restructure their strategic investment plans, thereby causing shifts in investment flows (Hossain et al., 2024). Theoretically, an increase in geopolitical risk is therefore expected to have a negative effect on FDI inflows.

Hypothesis 1: Geopolitical risks negatively affect FDI inflows to Türkiye.

On the other hand, economic policy uncertainty emerges as another key determinant that directly influences investment decisions. An uncertain policy environment weakens investors' expectations regarding future revenues, costs, and returns. It also increases the tendency to postpone or completely cancel investment projects, particularly those with a high degree of irreversibility. This situation reduces both domestic and foreign investment, leading to lower production and a decline in capital stock. In other words, uncertainty undermines the predictability of the host country environment and makes the expected returns on investment more risky (Zhang et al., 2023). Therefore, increases in economic policy uncertainty are theoretically expected to exert a deterrent effect on FDI inflows.

Hypothesis 2: Economic policy uncertainty negatively affects FDI inflows to Türkiye.

From both hypotheses, the main expectation is that risk and uncertainty conditions adversely affect FDI. The strength of this relationship, however, may differ across economies according to their stage of economic development. The literature generally agrees that developing countries are more exposed to these risk factors compared to developed economies (Yu and Wang, 2023).

3. Literature Review

3.1. Geopolitical Risk vs. Foreign Direct Investment

The relationship between geopolitical risk and FDI has become an increasingly important area of research in the international economics literature, particularly in recent years characterized by accelerating globalization and rising economic uncertainty. The main premise of this literature is that investment decisions are not determined solely by economic fundamentals but are also shaped by political instability, geopolitical tensions, and global uncertainties. As is well known, geopolitical risks create uncertainty in the investment environment, which may lead firms to postpone or completely cancel investment decisions. In this context, the literature examines the impact of geopolitical risk on FDI through empirical evidence that varies across country groups and over time.

The existing body of research largely indicates that geopolitical risk tends to discourage FDI inflows by increasing uncertainty and weakening investor confidence. Nguyen et al. (2022) show that geopolitical risk negatively affects FDI inflows in 18 developing economies and that this effect remains relatively stable over time. Similarly, Yu and Wang (2023), in their analysis of 41 countries, find that geopolitical risk generally reduces FDI inflows, although the magnitude of this effect varies depending on the level of economic development and becomes statistically insignificant in developed economies. Bussy and Zheng (2023) explain this relationship through firm-level behavior and institutional characteristics, emphasizing that governance quality, information accessibility, and technological intensity play a crucial role in determining responses to risk. They further suggest that investments characterized by strong governance and high research and development intensity are more resilient to risk, while multinational firms with better access to information tend to delay investment decisions when risk increases. In addition, studies such as Hossain et al. (2024) and Kapopoulos et al. (2025) indicate that geopolitical risk weakens FDI flows in the long run and that both global and country-specific risk components jointly influence investment decisions.

On the other hand, Truong et al. (2024) highlight that the effect of geopolitical risk on FDI varies depending on the time horizon. They find that increases in geopolitical risk may have a positive delayed effect on FDI in the short run, while the long-run effect is negative. Da Cunha et al. (2026) provide a notable contribution by examining the effects of US- and China-related geopolitical risks on FDI inflows to Latin America. Their findings show that US-related geopolitical risk has a negative effect on capital flows, whereas China-related geopolitical risk has a positive effect in the same period. The authors argue that the power rivalry between the United States and China reshapes investment flows toward developing regions. Al Mustofa et al. (2025) further show that this relationship varies across income groups, with middle-income countries being more adversely affected by geopolitical risk. However, strong institutions and developed financial systems are found to mitigate these negative effects.

In the case of Türkiye, existing studies are largely consistent with global findings. Afşar et al. (2022) show that geopolitical risk has a negative and statistically significant effect on FDI inflows to Türkiye. Similarly, Altıner and Bozkurt (2023) find that increasing risk levels adversely affect investment decisions. Özşahin et al. (2025) emphasize, by incorporating institutional factors, that geopolitical uncertainty reduces foreign investors' willingness to invest in Türkiye. Kalmaz and Adebayo (2024), using

different time and frequency dimensions, also confirm the negative impact of geopolitical risk on FDI flows, while Ünlü (2025) highlights that both geopolitical risk and economic growth are important determinants of investment inflows. Consistent with these findings, studies on Türkiye suggest that geopolitical risk is one of the key factors weakening the attractiveness of the investment environment. Table 1 summarizes the studies focusing on the relationship between geopolitical risk and FDI.

Table 1. Summary of Literature: Geopolitical Risk and FDI

Authors	Countries/Periods	Methods	Results
Cross-Country Evidence			
Nguyen et al. (2022)	18 Emerging Economies 1985-2019	Granger causality	GPR reduces FDI inflows.
Bussy and Zheng (2023)	18 Emerging Economies 2003-2019	Fixed effect	GPR reduces FDI inflows.
Yu and Wang (2023)	41 Countries 2003-2020	Fixed effect	GPR reduces FDI inflows.
Truong et al. (2024)	Vietnam 1986-2021	ARDL	GPR decreases FDI in the long run but increases it in the short run.
Yılmaz (2024)	10 high-risky countries 2000-2022	CS-ARDL	GPR reduces FDI inflows in both the short and long run.
Hossain et al. (2024)	5 Asian Countries 1996-2019	Panel cointegration	GPR reduces FDI inflows.
Al Mustofa et al. (2025)	42 Countries 1990-2020	Fixed effect GMM	GPR reduces FDI inflows, especially in middle income countries.
Kapopoulos et al. (2025)	43 Countries 1985-2022	Panel regression	GPR reduces FDI inflows.
Da Cunha et al. (2026)	7 Latin American Countries 2000-2022	Fixed effect	US GPR is negative for FDI inflows, Chinese GPR is positive.
Country-Specific Evidence: Türkiye			
Afşar et al. (2022)	Türkiye 1998-2018	ARDL	GPR reduces FDI inflows.
Özşahin et al. (2025)	Türkiye 2003-2020	ARDL	GPR reduces FDI inflows.
Altıner and Bozkurt (2023)	Türkiye 1985-2020	ARDL	GPR reduces FDI inflows.
Kalmaz and Adebayo (2024)	Türkiye 1988-2020	Wavelet Coherence	GPR reduces FDI inflows.
Ünlü (2025)	Türkiye 1985-2025	ARDL	GPR reduces FDI inflows.

Overall, the literature suggests that there is a predominantly negative relationship between geopolitical risk and FDI. However, the magnitude and direction of this effect vary depending on country-specific characteristics, institutional quality, the time horizon, and global power dynamics. In particular, developing countries and middle-income economies tend to be more vulnerable to geopolitical risk shocks, whereas strong institutional frameworks and well-developed financial systems can partially offset these adverse effects. Moreover, the literature highlights that the effects of geopolitical risk differ over time, and that global power competition—especially the US–China rivalry—plays a significant role in shaping investment flows. These aspects constitute important areas for future research.

3.2. Economic Policy Uncertainty vs. Foreign Direct Investment

In recent years, the link between economic policy uncertainty and FDI has attracted growing scholarly attention within the field of international economics. The acceleration of globalization, rising financial integration, and more frequent macroeconomic shocks have shown that investment decisions are shaped not only by economic fundamentals but also by the predictability of policy environments. In this context, economic policy uncertainty is considered an important factor that increases investors' risk perceptions in decision-making processes, particularly for long-term capital movements. The general tendency in the literature suggests that uncertainty weakens the investment environment

and reduces capital inflows, although the direction and magnitude of this effect may vary depending on country groups, the source of uncertainty, and institutional structures.

Empirical evidence from the international literature largely suggests that economic policy uncertainty discourages FDI flows by increasing risk perceptions and weakening investors' long-term decision-making incentives. Avom et al. (2020), using a sample of 138 countries, show that global economic policy uncertainty reduces FDI inflows overall, with a stronger effect in developing countries. Similarly, Choi et al. (2021) find that domestic policy uncertainty significantly reduces FDI inflows in OECD countries, particularly in economies with lower financial development. Zhang et al. (2023) and Tabash (2025) also confirm statistically significant negative effects of economic policy uncertainty on FDI flows in Asian and South Asian countries, respectively. In particular, Zhang et al. (2023) show that foreign investment is more sensitive to policy uncertainty compared to domestic investment, and that both local and global uncertainty reduce investment inflows. However, Canh et al. (2020) find that while domestic uncertainty reduces FDI inflows, global uncertainty measures may, in some cases, have a positive effect depending on the index used (such as the world uncertainty index). This suggests that investors may respond heterogeneously to different types of uncertainty. Similarly, Suleyman (2020) shows that domestic uncertainty does not always have a statistically significant effect in some regions, while European and global uncertainty measures reduce FDI inflows. More recent studies, such as Phan et al. (2024), confirm the overall negative impact of economic policy uncertainty on FDI, while also showing that the effect varies across investment types. In particular, cross-border mergers and acquisitions appear to be more resilient during periods of uncertainty. Odionye et al. (2026), focusing on Sub-Saharan African countries, find that domestic economic policy uncertainty reduces FDI inflows, whereas global uncertainty may increase FDI under certain conditions. They also emphasize that institutional quality plays a moderating role in this relationship.

For Türkiye, direct evidence on the relationship between economic policy uncertainty and FDI remains relatively scarce, although existing studies offer useful complementary insights through related macroeconomic and investment channels. Sahinoz and Cosar (2018) show that economic policy uncertainty in Türkiye increases particularly during crisis and election periods, and that this rise negatively affects economic activity through the investment channel. Tiryaki and Tiryaki (2018) find that global economic policy uncertainty influences economic indicators through financial markets and has a negative effect on stock returns. At the firm level, Azimli (2022), using data from Türkiye and four other countries, examines the relationship between economic policy uncertainty and corporate capital investment. The results show no statistically significant relationship, suggesting that the real options mechanism—where firms delay investment under uncertainty—may not hold in these countries. Finally, Türkmen and Yarbaşı (2023) find that FDI in Türkiye is affected by uncertainty in a sector-specific manner, with significant effects observed in the industrial sector but not in the services sector. These findings suggest that policy-related uncertainty affects investment behavior in Türkiye in a way that is neither direct nor homogeneous, but instead varies across sectors and contexts.

Table 2. Summary of Literature: Economic Policy Uncertainty and FDI

Authors	Countries/Periods	Methods	Results
Cross-Country Evidence			
Avom et al. (2020)	138 Countries 1996-2018	GMM	EPU reduces FDI inflows, especially in developing and emerging countries.
Canh et al. (2020)	21 Countries 2003-2013	GMM	EPU reduces FDI inflows, while WUI increases it.
Suleyman (2020)	10 Central and Eastern Europe Countries 1990-2018	Fixed Effect Random Effect	Domestic EPU has an insignificant effect on FDI, while global EPU reduces it.
Choi et al. (2021)	16 OECD Countries 1985-2013	Fixed Effect	EPU reduces FDI inflows, especially in countries with less financial development.
Zhang et al. (2023)	48 Asian Countries 1995-2020	GMM	EPU reduces FDI inflows.
Phan et al. (2024)	213 Countries 2003-2020	Fixed Effect	EPU reduces FDI inflows.
Tabash (2025)	South Asian Region 2000-2019	GMM	EPU reduces FDI inflows.
Odionye et al. (2026)	23 SSA Countries 2004-2022	Panel Quantile MMQR	EPU reduces FDI inflows.
Country-Specific Evidence: Türkiye			
Sahinoz and Cosar (2018)	Türkiye 1998-2014	VAR	EPU reduces real investments.
Tiryaki and Tiryaki (2018)	Türkiye 1997-2018	ARDL	EPU reduces stock returns.
Azimli (2022)	5 Countries (Including Türkiye) 2000-2018	Fixed Effect	EPU is not significantly related to corporate capital investment.
Türkmen and Yarbaşı (2023)	Türkiye 2005-2022	ARDL	EPU significantly affects manufacturing, but not services.

Table 2 presents a synthesis of empirical studies examining how economic policy uncertainty shapes FDI. Taken together, the existing evidence suggests that policy-related uncertainty generally discourages foreign investment. Nevertheless, the strength and even the direction of this effect may differ according to whether uncertainty originates from domestic or global sources, as well as countries' development levels, financial depth, and institutional quality. Moreover, some studies suggest that under certain conditions, global uncertainty may even increase investment flows or produce heterogeneous effects across sectors. The case of Türkiye further indicates that this relationship operates through more complex and indirect channels, with sectoral differences and macroeconomic conditions playing a decisive role.

4. Data

To capture the role of geopolitical risk and policy-related uncertainty in shaping FDI inflows to Türkiye, the empirical model specified in Equation (1) is employed.

$$\ln FDI_t = \alpha + \beta_1 \ln EPU_t + \beta_2 \ln GPR_t + \beta_3 \ln GDP_t + \beta_4 \ln INQ_t + \varepsilon_t \quad (1)$$

The dependent variable, FDI, is one of the key components of the economic globalization process and represents FDI inflows. The main explanatory variables, EPU and GPR, are included as the core sources of risk and uncertainty in the analysis, representing economic policy uncertainty and geopolitical risk, respectively. GDP captures economic growth and is measured as real gross domestic product per capita. Finally, INQ represents the level of institutionalization and is included as a proxy for institutional quality. Detailed information on all variables is provided in Table 3.

Table 3. Variable Definitions

Symbol	Description	Measurement	Source
FDI	FDI Inflows	FDI, net inflows (% of GDP)	WDI
EPU	Economic Uncertainty	World uncertainty index	FRED
GPR	Geopolitical Risk	Geopolitical risk index	Caldara and Iacoivello (2022)
GDP	Economic Growth	GDP (constant 2015 US$)	WDI
INQ	Institutionalization	Institutional quality	ICRG

Note: ICRG: The International Country Risk Guide. **FRED:** Federal Reserve Economic Data. **WDI:** World Development Indicator.

In Equation (1), α denotes the constant term, while ε represents the error term. The coefficients β_1, β_2, β_3 and β_4 indicate the elasticity of economic policy uncertainty, geopolitical risk, economic growth, and institutional quality, respectively. All variables are transformed into logarithmic form prior to estimation. Finally, the sample period is determined primarily by the availability of the geopolitical risk series, which begins in 1985. Accordingly, the study covers the period from 1985 to 2024.

5. Econometric Methodology

The econometric analysis is conducted through a four-step procedure. First, the order of integration of the variables is assessed by applying the conventional Augmented Dickey-Fuller (ADF) test together with the Fourier ADF unit root test, which accounts for smooth structural shifts through fractional frequency values. Second, the presence of a long-run equilibrium relationship among the variables is investigated using the Fourier ADL cointegration approach. Third, after identifying the cointegration structure, the long-run coefficients are estimated through the Dynamic Ordinary Least Squares (DOLS) estimator. In the final step, the direction of causal linkages among the variables is examined by employing the Fourier Toda-Yamamoto causality test.

5.1. Fourier ADF Unit Root Test

The Fourier ADF unit root testing approach proposed by Christopoulos and Leon-Ledesma (2011) accounts for structural changes by incorporating fractional frequency components. In their framework, integer frequency values are associated with transitory structural changes, whereas fractional frequency values capture more persistent shifts in the deterministic path of the series. Accordingly, a stochastic process $y_t$may be defined with a time-varying deterministic component $d(t)$and an error term $v_t \sim N(0, \sigma)$, as shown in Equation (2):

$$y_t = d(t) + v_t \tag{2}$$

For the deterministic component $d(t)$, the Fourier representation given in Equation (3) is used:

$$d(t) = \gamma_0 + \gamma_1 \sin\left(\frac{2\pi kt}{T}\right) + \gamma_2 \cos\left(\frac{2\pi kt}{T}\right) \tag{3}$$

In the Fourier specification in Equation (3), k denotes the frequency, t represents the time trend, and T indicates the sample size. In the first stage of the Fourier ADF test with fractional frequencies, the optimal frequency value (k^*) is determined. For this purpose, Equation (3) is estimated using Ordinary Least Squares (OLS) for each fractional value of k in the range $0.1 \leq k \leq 5$, with increments of 0.1. The value of k that minimizes the sum of squared residuals is selected as k^*. Subsequently, the OLS residuals of the model are obtained as:

$$\hat{v}_t = y_t - \hat{\gamma}_0 + \hat{\gamma}_1 \sin\left(\frac{2\pi k^* t}{T}\right) + \hat{\gamma}_2 \cos\left(\frac{2\pi k^* t}{T}\right) \tag{4}$$

In the second stage of the test, a unit root test is applied to the obtained residuals. For this purpose, the standard ADF regression given in Equation (5) is estimated:

$$\Delta v_t = \alpha_1 v_{t-1} + \sum_{j=1}^{p} \beta_j \Delta v_{t-j} + u_t \tag{5}$$

The null hypothesis of a unit root, $H_0: \alpha_1 = 0$, is tested. In the third and final stage of the Fourier ADF test, the presence of unknown smooth structural breaks in the data generating process of y_t is examined using the null hypothesis $H_0: \gamma_1 = \gamma_2 = 0$. The F-statistic is used to test this null hypothesis.

5.2. Fourier ADL Cointegration Test

The Fourier ADL cointegration approach introduced by Banerjee et al. (2017) extends the autoregressive distributed lag framework by integrating a Fourier function into the model specification. The Fourier ADL approach helps to avoid the loss of power that typically arises in cointegration tests when a large number of dummy variables are used to capture structural breaks (Banerjee et al., 2017). In the Fourier ADL procedure, the conditional model given in Equation (6) is employed:

$$\Delta y_{1t} = d(t) + \delta_1 y_{1,t-1} + \gamma' y_{2,t-1} + \varphi' \Delta y_{2t} + \epsilon_t \quad (6)$$

where, γ, φ ve y_{2t} are n×1 parameter vectors and the explanatory variables. The term . $d(t)$, denotes the deterministic component specified through the Fourier function. In the first stage of the Fourier ADL cointegration test, Equation (6) is estimated, and the optimal frequency value k^*, which minimizes the sum of squared residuals, is determined. Following Ilkay et al. (2021), and in line with Christopoulos and Leon-Ledesma (2011), fractional frequency values are considered in the range $0.1 \leq k \leq 5$ with increments of 0.1 in order to capture permanent structural changes. In the Fourier ADL cointegration framework, the null hypothesis of no cointegration is tested as $H_0: \delta_1 = 0$. To test this hypothesis, the test statistic given in Equation (7) is used:

$$t_{ADL}^{F} = \frac{\hat{\delta}_1}{se(\hat{\delta}_1)} \quad (7)$$

where $\hat{\delta}_1$ denotes the OLS estimator of δ_1 in Equation (6), while $se(\hat{\delta}_1)$ represents its standard error.

5.3. Fourier Toda-Yamamoto Causality Test

The Toda and Yamamoto (1995) causality framework enables the examination of causal linkages among variables regardless of whether they are integrated at different orders. However, the standard Toda–Yamamoto approach does not account for structural breaks. To address this limitation, Nazlioglu et al. (2016) propose a Fourier causality test that builds on the Toda–Yamamoto framework and allows for structural breaks in the form of gradual and smooth changes.

In the Toda–Yamamoto approach, the VAR model specified in Equation (8) is estimated:

$$y_t = \alpha + \beta_1 y_{t-1} + \cdots + \beta_{p+d} y_{t-(p+d)} + \varepsilon_t \quad (8)$$

where, p denotes the optimal lag length, while d represents the maximum order of integration of the variables. The term ε_t refers to the residuals, which are assumed to follow a white noise process. In the Fourier Toda–Yamamoto causality framework, the model is extended as shown in Equation (9):

$$y_t = \alpha_0 + \gamma_1 sin\left(\frac{2\pi kt}{T}\right) + \gamma_2 cos\left(\frac{2\pi kt}{T}\right) + \beta_1 y_{t-1} + \cdots + \beta_{p+d} y_{t-(p+d)} + \varepsilon_t \quad (9)$$

In the Fourier Toda–Yamamoto test, the null hypothesis of no causality ($H_0: \beta_1 = \cdots = \beta_p = 0$) is examined using the Wald test statistic.

6. Empirical Findings

Prior to the investigation of a long-run equilibrium relationship, the time-series properties of the variables are first assessed. For this purpose, the study applies the conventional ADF unit root test together with the Fourier ADF (FADF) test of Christopoulos and Leon-Ledesma (2011), which captures smooth structural shifts through fractional frequency components. The corresponding unit root test results are presented in Table 4.

Table 4. Unit Root Test Results

Variables	ADF	Min SSR	Frequency	Fourier ADF	F-statistic
FDI	-2.438	8.915	0.7	-3.563	40.435*
ΔFDI	-6.908*	8.569	5	-5.569*	1.188
EPU	-1.571	15.302	0.1	-3.411	18.494*
ΔEPU	-5.998*	16.169	3.3	-6.912*	1.435
GPR	-0.997	8.415	0.5	-3.443	16.783*
ΔGPR	-8.591*	12.651	3.6	-6.425*	0.697
GDP	-2.034	0.067	0.2	-3.438	1519.836*
ΔGDP	-4.166**	0.055	4.5	-2.854	4.091
INQ	-0.874	0.516	0.8	-3.042	8.943*
ΔINQ	-4.366*	0.355	4.9	-3.601**	2.539

Note: ***, **, and * symbols signify significance at the levels of 10%, 5%, and 1%.

The results reported in Table 4 show that the Fourier term is significant for all variables in their level specifications. Accordingly, when the Fourier component is significant, the order of integration is determined primarily on the basis of the Fourier ADF test, whereas the conventional ADF test is used where the Fourier component is not supported. The integration order is determined through a combined assessment of the unit root test results. The findings indicate that none of the variables is stationary in levels. However, all series become stationary after first differencing. Therefore, the unit root findings confirm that all variables follow an I(1) process, providing an appropriate basis for implementing the Fourier ADL cointegration test.

Table 5. Cointegration Test Results

Min AIC	Frequency	Fourier ADL test statistic	Decision
0.685	1.6	-5.054**	Cointegration
	Critical Values		
Fourier ADL	%1	%5	%10
	-5.314	-4.626	-4.261

Note: ***, **, and * symbols signify significance at the levels of 10%, 5%, and 1%.

Table 5 presents the results of the Fourier ADL cointegration test with fractional frequency components. First, the optimal frequency value is estimated as a fractional value (1.6), thereby capturing permanent structural breaks in the model. The Fourier ADL test statistic is found to be -5.054. Since this value exceeds the 5% critical value in absolute terms, the null hypothesis of no cointegration is rejected. This result indicates the existence of a cointegration relationship among the variables. Accordingly, the variables move together in the long run, implying that any long-run analysis based on their level values will not suffer from spurious regression problems.

The long-run parameters are obtained through the DOLS estimator proposed by Stock and Watson (1993). The DOLS estimator is preferred because it provides consistent long-run estimates in the presence of endogeneity and serial correlation in cointegrated systems (Stock and Watson, 1993). The estimation outcomes are presented in Table 6, with the model specification incorporating trigonometric components to account for smooth structural changes.

Table 6. Long-run Coefficients

Variables	Coefficients	Std. Error	t-statistic	p-value
EPU	-0.229*	0.062	-3.665	0.003
GPR	-0.265**	0.114	-2.333	0.036
GDP	2.645*	0.076	34.772	0.000
INQ	2.616*	0.168	15.538	0.000
C	-22.955*	0.855	-26.849	0.000
Sin	-0.130*	0.024	-5.511	0.000
Cos	0.386*	0.014	28.445	0.000
Adjusted R-squared	0.742	S.E. of regression	0.411	

Note: ***, **, and * symbols signify significance at the levels of 10%, 5%, and 1%.

The results indicate that the coefficients of both economic policy uncertainty and geopolitical risk are negative and statistically significant. In this context, the estimated

coefficients for economic policy uncertainty and geopolitical risk are -0.229 and -0.265, respectively. These findings confirm both Hypothesis 1 and Hypothesis 2 tested in the study. Accordingly, Hypothesis 1, which states that geopolitical risk negatively affects FDI inflows to Türkiye, is empirically supported. This result is consistent with previous studies such as Nguyen et al. (2022), Yu and Wang (2023), and Hossain et al. (2024), as well as Türkiye-specific evidence provided by Afşar et al. (2022), Altıner and Bozkurt (2023), and Ünlü (2025). These studies similarly emphasize that geopolitical risk increases uncertainty in the investment environment and reduces capital inflows.

In a similar way, Hypothesis 2, which suggests that economic policy uncertainty negatively affects FDI inflows to Türkiye, is also confirmed. This finding is in line with the results of Avom et al. (2020), Choi et al. (2021), and Zhang et al. (2023), as well as country-specific evidence for Türkiye provided by Sahinoz and Cosar (2018) and Türkmen and Yarbaşı (2023). Overall, the results are consistent both with the general literature and with empirical evidence specific to Türkiye. In this context, a 1% increase in economic policy uncertainty and geopolitical risk reduces FDI inflows by 0.229% and 0.265%, respectively. This finding reflects the vulnerability of Türkiye as a developing economy to risk and uncertainty shocks.

On the other hand, the coefficients of economic growth and institutional quality are found to be positive and statistically significant. These results indicate that a 1% increase in economic growth and institutional quality increases FDI inflows by 2.645% and 2.616%, respectively, in the long run.

The results also allow for a clearer interpretation of the individual and joint effects of economic policy uncertainty, geopolitical risk, and institutional quality on FDI inflows. When considered individually within the long-run framework, economic policy uncertainty exerts a negative effect on FDI, indicating that a less predictable policy environment discourages foreign investors by increasing perceived investment risk. Similarly, geopolitical risk has a negative effect on FDI inflows, suggesting that regional conflicts, political tensions, and security-related concerns weaken Türkiye's attractiveness as an investment destination. In contrast, institutional quality has a positive effect on FDI, implying that stronger institutions, regulatory reliability, and the rule of law improve the predictability of the investment environment and support foreign capital inflows.

When these variables are evaluated together, the findings suggest that FDI inflows to Türkiye are shaped by the interaction of risk, uncertainty, and institutional conditions. While economic policy uncertainty and geopolitical risk create a deterrent effect on foreign investors, institutional quality functions as a stabilizing factor that can partially offset the adverse effects of uncertainty and risk. Therefore, the joint interpretation of the results indicates that attracting FDI requires not only reducing policy uncertainty and geopolitical tensions, but also strengthening institutional credibility and regulatory predictability.

Table 7. Causality Analysis

Causality path	Frequency	WALD-statistic	p-value
EPU → FDI	0.8	4.362**	0.044
GPR → FDI	0.8	0.340	0.532
GDP → FDI	0.2	6.660**	0.012
INQ → FDI	0.7	0.001	0.990
FDI → EPU	0.8	0.036	0.851
FDI → GPR	0.8	0.004	0.949
FDI → GDP	0.2	0.860	0.374
FDI → INQ	0.7	0.588	0.443

Note: ***, **, and * symbols signify significance at the levels of 10%, 5%, and 1%.

In the final stage of the econometric analysis, the Fourier Toda–Yamamoto causality test with fractional frequency components is employed to check the robustness of the cointegration results. Table 7 reports the causal relationships between FDI and economic

policy uncertainty, geopolitical risk, economic growth, and institutional quality. The results of the Fourier Toda–Yamamoto causality test indicate the existence of a unidirectional causality running from economic policy uncertainty and economic growth to FDI. The absence of a significant causal relationship from geopolitical risk to FDI does not contradict the long-run DOLS findings. Instead, it indicates that the effect of geopolitical risk is primarily reflected in the long-run equilibrium relationship rather than through direct causal linkages.

7. Discussion

The findings show that economic policy uncertainty and geopolitical risk both discourage FDI inflows to Türkiye. The negative and statistically significant coefficients suggest that uncertainty and risk factors directly influence investment decisions in developing economies with a high dependence on external financing, such as Türkiye. In particular, the adverse impact of economic policy uncertainty highlights the importance that investors attach to predictability and policy stability. Fluctuations in monetary, fiscal, and regulatory policies observed in Türkiye from time to time may distort the expected risk–return balance for foreign investors, thereby reducing investment appetite. Similarly, the negative effect of geopolitical risk underscores the importance of Türkiye's geographical position and its sensitivity to regional political developments in shaping investment decisions. Political and military tensions in regions such as the Middle East, the Balkans, and the Eastern Mediterranean are perceived by international investors as an additional risk premium, thereby constraining capital inflows. This finding suggests that geopolitical stability is a critical determinant of FDI attraction.

Conversely, the positive and statistically significant coefficient of economic growth suggests that stronger growth performance enhances Türkiye's attractiveness for foreign investors. By expanding market potential and improving expected returns, economic growth creates more favorable conditions for FDI inflows. This result is also consistent with the market size hypothesis and suggests that Türkiye's dynamic domestic demand structure supports investment decisions. The positive effect of institutional quality is particularly noteworthy. The results show that factors such as the rule of law, regulatory effectiveness, and institutional reliability play a decisive role in attracting FDI to Türkiye. Improvements in institutional quality reduce transaction costs, enhance predictability in the investment environment, and facilitate long-term decision-making for foreign investors. In this context, strengthening institutional structures clearly enhances Türkiye's capacity to attract foreign investment.

Overall, the findings suggest that FDI in Türkiye is shaped both by economic fundamentals (growth and institutional quality) and by risk and uncertainty factors (economic policy uncertainty and geopolitical risk). This dual structure implies a two-dimensional policy strategy for policymakers: on the one hand, strengthening macroeconomic stability and institutional quality, and on the other, implementing structural measures to mitigate the adverse risk and uncertainty pressures that weaken the investment climate.

8. Conclusion and Policy Recommendation

This study examines the long-run effects of geopolitical risk and economic policy uncertainty on FDI inflows to Türkiye over the period 1985–2024. In addition to these two risk-related variables, economic growth and institutional quality are included in the empirical model in order to control for the broader macroeconomic and institutional environment. By employing Fourier-based time-series techniques that allow for smooth structural changes, the study provides evidence on how uncertainty, geopolitical tensions, growth dynamics, and institutional conditions shape foreign investment inflows in Türkiye. The empirical findings indicate that both economic policy uncertainty and geopolitical risk have negative and statistically significant effects on FDI inflows.

Specifically, a 1% increase in economic policy uncertainty reduces FDI inflows by 0.229%, while a 1% increase in geopolitical risk reduces FDI inflows by 0.265% in the long run. These results suggest that foreign investors are highly sensitive to uncertainty and risk conditions in Türkiye. Economic policy uncertainty may weaken investor confidence by increasing concerns about future monetary, fiscal, and regulatory decisions. Similarly, geopolitical risk may raise the perceived risk premium of investing in Türkiye due to regional conflicts, security concerns, and political tensions in surrounding regions. Therefore, the findings confirm that FDI inflows to Türkiye are shaped not only by economic fundamentals but also by the stability and predictability of the policy and geopolitical environment.

The results also show that economic growth and institutional quality have positive and statistically significant effects on FDI inflows. The positive coefficient of economic growth implies that a larger and expanding market increases Türkiye's attractiveness for foreign investors by improving expected returns and market opportunities. Likewise, the positive effect of institutional quality indicates that stronger institutions, rule of law, regulatory reliability, and policy credibility reduce transaction costs and support long-term investment decisions. The causality analysis further supports the importance of policy uncertainty and economic growth, as unidirectional causality is found from economic policy uncertainty and economic growth to FDI. Taken together, these findings imply that Türkiye's ability to attract foreign capital depends on a combination of reduced uncertainty, lower geopolitical risk, stronger institutional quality, and sustained economic performance.

Based on these findings, several policy recommendations can be proposed. First, reducing economic policy uncertainty should be considered a central priority for improving Türkiye's investment climate. For this purpose, monetary, fiscal, and regulatory policies should be designed within a more coordinated, transparent, and predictable framework. Sudden and frequent changes in interest rate policy, taxation, investment regulations, and sectoral incentives may increase perceived uncertainty among foreign investors. Therefore, policymakers should strengthen policy communication, announce medium-term policy targets more clearly, and ensure consistency between announced objectives and implemented policies.

Second, the adverse effect of geopolitical risk suggests that foreign investment policy cannot be separated from foreign policy and regional stability. Türkiye should strengthen diplomatic channels, reduce the economic costs of regional tensions, and develop mechanisms that protect economic relations from political shocks as much as possible. In this context, investment protection agreements, stronger dispute-resolution mechanisms, and clearer guarantees for foreign investors may help reduce the risk premium associated with geopolitical uncertainty. Moreover, improving Türkiye's image as a stable and reliable investment destination can contribute to sustaining FDI inflows even during periods of regional instability.

Third, institutional quality should be strengthened as a complementary policy tool for attracting FDI. Improvements in rule of law, judicial efficiency, transparency, bureaucratic quality, regulatory stability, and protection of property rights may reduce uncertainty-driven pressures on foreign investors. In particular, simplifying administrative procedures, ensuring equal treatment of domestic and foreign investors, and increasing the predictability of investment-related regulations can make Türkiye's investment environment more credible. These institutional improvements may also moderate the negative effects of policy uncertainty and geopolitical risk.

Fourth, sustaining economic growth remains important for attracting market-seeking and efficiency-seeking FDI. However, growth alone may not be sufficient if it is accompanied by high uncertainty, weak institutions, or geopolitical tensions. Therefore, growth-oriented policies should be supported by macroeconomic stability, price stability, exchange rate predictability, and structural reforms that improve productivity. Policies that encourage high-value-added production, technology-intensive

sectors, and stronger integration into global value chains may also increase the quality and durability of FDI inflows.

Overall, the findings indicate that Türkiye's FDI performance depends on a balanced policy framework that simultaneously reduces uncertainty, mitigates geopolitical risk, strengthens institutional quality, and supports sustainable economic growth. The results suggest that policies aimed only at increasing market size or providing investment incentives may remain insufficient unless they are supported by a stable policy environment and credible institutions. Therefore, a comprehensive investment strategy for Türkiye should combine macroeconomic stability, transparent policymaking, institutional reform, diplomatic risk management, and sector-specific investment incentives.

This study is subject to some limitations. Since the analysis focuses only on Türkiye, the findings may not be directly generalizable to other emerging economies. In addition, the study relies on selected indicators of geopolitical risk, economic policy uncertainty, and institutional quality. Future research may extend the analysis by using sectoral FDI data, alternative uncertainty indicators, or comparative country samples. Such extensions may provide further insights into whether the effects of uncertainty and geopolitical risk differ across sectors, regions, or country groups.

References

Afşar, M., Doğan, E., & Doğan, B. Ö. (2022). Does higher geopolitical risk limits Turkish foreign direct investments?. *Journal of Mehmet Akif Ersoy University Economics and Administrative Sciences Faculty*, *8*(3), 1456-1475. https://doi.org/10.30798/makuiibf.859137

Al Mustofa, M. U., & Risyad, M. H. (2025). Geopolitical risk and foreign direct investment: the role of liberalization, institutions and financial markets. Studies in Economics and Finance, 42(4), 730-758. https://doi.org/10.1108/SEF-09-2024-0621

Altıner, A., & Bozkurt, E. (2023). Reflections of geopolitical risk on foreign direct investments: The case of Türkiye. *İnsan ve Toplum Bilimleri Araştırmaları Dergisi*, *12*(3), 1292-1309. https://doi.org/10.15869/itobiad.1271884

Avom, D., Njangang, H., & Nawo, L. (2020). World economic policy uncertainty and foreign direct investment. *Economics Bulletin*, *40*(2), 1457-1464.

Azimli, A. (2022). The impact of policy, political and economic uncertainty on corporate capital investment in the emerging markets of Eastern Europe and Turkey. *Economic Systems*, *46*(2), 100974. https://doi.org/10.1016/j.ecosys.2022.100974

Banerjee, P., Arčabić, V., & Lee, H. (2017). Fourier ADL cointegration test to approximate smooth breaks with new evidence from crude oil market. *Economic Modelling*, *67*, 114-124. https://doi.org/10.1016/j.econmod.2016.11.004

Bussy, A., & Zheng, H. (2023). Responses of FDI to geopolitical risks: The role of governance, information, and technology. *International Business Review*, *32*(4), 102136. https://doi.org/10.1016/j.ibusrev.2023.102136

Caldara, D. & Iacoviello, M. (2022). Measuring geopolitical risk, *American Economic Review*, *112*(4), 1194-1225. https://www.matteoiacoviello.com/gpr.htm

Canh, N. P., Binh, N. T., Thanh, S. D., & Schinckus, C. (2020). Determinants of foreign direct investment inflows: The role of economic policy uncertainty. *International Economics*, *161*, 159-172. https://doi.org/10.1016/j.inteco.2019.11.012

Choi, S., Furceri, D., & Yoon, C. (2021). Policy uncertainty and foreign direct investment. *Review of International Economics*, *29*(2), 195-227. https://doi.org/10.1111/roie.12495

Christopoulos, D. K., & Leon-Ledesma, M. A. (2011). International output convergence, breaks, and asymmetric adjustment. *Studies in Nonlinear Dynamics & Econometrics*, *15*(3), 1-31.

Da Cunha, H. C., Singh, V., Amal, M., Farrell, C., Mangal, S., & Grellmann, G. (2026). The impact of US and China geopolitical risk on foreign direct investment in Latin America. *International Business Review*, *35*(1), 102533. https://doi.org/10.1016/j.ibusrev.2025.102533

Dixit, A. K., & Pindyck, R. S. (1994). *Investment under uncertainty*. Princeton university press.

Hossain, M. S., Voumik, L. C., Ahmed, T. T., Alam, M. B., & Tasmim, Z. (2024). Impact of geopolitical risk, GDP, inflation, interest rate, and trade openness on foreign direct investment: Evidence from five Southeast Asian countries. *Regional Sustainability*, *5*(4), 100177. https://doi.org/10.1016/j.regsus.2024.100177

Ilkay, S. C., Yilanci, V., Ulucak, R., & Jones, K. (2021). Technology spillovers and sustainable environment: Evidence from time-series analyses with Fourier extension. *Journal of Environmental Management*, *294*, 113033. https://doi.org/10.1016/j.jenvman.2021.113033

Kalmaz, D. B., & Adebayo, T. S. (2024). Understanding the factors that influence foreign direct investment in Turkey: Exploring the impact of country risks and geopolitical risk. *Journal of International Commerce, Economics and Policy*, *15*(02), 2450014. https://doi.org/10.1142/S1793993324500145

Kızılkaya, F., Kızılkaya, O., & Mike, F. (2026). Does geopolitical risk escalate environmental degradation in Turkey? Evidence from a Fourier approach. *Environment, Development and Sustainability*, *28*(3), 7085-7106.

Kapopoulos, T., Sakkas, A., & Drakos, K. (2025). Geopolitical risk and foreign direct investment inflows: the moderating role of water and energy risks. *International Journal of Finance & Economics*, *30*(4), 4039-4062. https://doi.org/10.1002/ijfe.3106

Nazlioglu, S., Gormus, N. A., & Soytas, U. (2016). Oil prices and real estate investment trusts (REITs): Gradual-shift causality and volatility transmission analysis. *Energy Economics*, *60*, 168-175. https://doi.org/10.1016/j.eneco.2016.09.009

Nguyen, T. T. T., Pham, B. T., & Sala, H. (2022). Being an emerging economy: to what extent do geopolitical risks hamper technology and FDI inflows?. *Economic Analysis and Policy*, *74*, 728-746. https://doi.org/10.1016/j.eap.2022.04.005

Odionye, J. C., Duru, I. U., Nwaeze, N. C., & Agoh, N. (2026). Heterogeneous Influence of Economic Policy Uncertainty on Foreign Direct Investment in Sub-Saharan Afrıcan (SSA) Countries: Moderating Role of Institutional Quality. *Global Economy Journal*, 1-27. https://doi.org/10.1142/S2194565926500028

Özşahin, Ş., Üçler, G., & Uysal, D. (2022). Jeopolitik risk ve yolsuzluğun doğrudan yabancı yatırımlar için önemi: Türkiye için ampirik bulgular. *Bingöl Üniversitesi İktisadi ve İdari Bilimler Fakültesi Dergisi*, *6*(2), 257-286. https://doi.org/10.33399/biibfad.1120293

Phan, N. H., Bui, V. D., & Nguyen, L. T. Q. (2024). How economic policy uncertainty affects foreign direct investment inflows: implications for the post-Covid-19 pandemic period. *International Journal of Social Economics*, *51*(2), 282-296. https://doi.org/10.1108/IJSE-02-2023-0114

Quazi, R. (2007). Economic freedom and foreign direct investment in East Asia. *Journal of the Asia Pacific Economy*, *12*(3), 329-344. https://doi.org/10.1080/13547860701405755

Sahinoz, S., & Erdogan Cosar, E. (2018). Economic policy uncertainty and economic activity in Turkey. *Applied Economics Letters*, *25*(21), 1517-1520. https://doi.org/10.1080/13504851.2018.1430321

Stock, J. H., & Watson, M. W. (1993). A simple estimator of cointegrating vectors in higher order integrated systems. *Econometrica: journal of the Econometric Society*, *61*(4), 783-820. https://doi.org/10.2307/2951763

Suleyman, S. (2020). How Economic Uncertainty affects foreign direct investment inflows: Evidence from Central and Eastern European countries. *Ekonomski vjesnik/Econviews-Review of Contemporary Business, Entrepreneurship and Economic Issues*, *33*(2), 451-562.

Tabash, M. I. (2025). Economic policy uncertainty and foreign direct investment inflow: The role of institutional quality in South Asia region. *Research in International Business and Finance*, *76*, 102860. https://doi.org/10.1016/j.ribaf.2025.102860

Tiryaki, A., & Tiryaki, H. N. (2018). Determinants of Stock Returns in Turkey under the Impacts of Capital Inflows and Global Economic Policy Uncertainty. *Journal of Current Researches on Social Sciences (JoCReSS)*, *8*(4), 413-436.

Toda, H. Y., & Yamamoto, T. (1995). Statistical inference in vector autoregressions with possibly integrated processes. *Journal of Econometrics*, *66*(1-2), 225-250. https://doi.org/10.1016/0304-4076(94)01616-8

Truong, L. D., Friday, H. S., & Pham, T. D. (2024). The effects of geopolitical risk on foreign direct investment in a transition economy: Evidence from Vietnam. *Journal of Risk and Financial Management*, *17*(3), 101. https://doi.org/10.3390/jrfm17030101

Türkmen, A., & Yarbaşı, İ. Y. (2023). Sektörel düzeyde doğrudan yabancı yatırımlar ile belirsizlik ilişkisi: Türkiye örneği. *Ekonomi ve Finans Politikaları Üzerine Çalışmalar*, içinde (105-129), Özgür Yayıncılık.

United Nations Conference on Trade and Development (2025). 2025 World Investment Report: International Investment in the Digital Economy. United Nations Publications, New York, US.

Ünlü, A. (2025). Türkiye'de doğrudan yabancı yatırımların makroekonomik ve jeopolitik parametrelerle şekillenmesi: ARDL perspektifinden bulgular. *Fiscaoeconomia*, *9*(3), 1323-1340.

World Bank (2026). The World Bank in Türkiye. Access date: 24 April 2026. https://www.worldbank.org/en/country/turkey/overview

Yılanci, V., Mike, F., Ursavaş, U., & Kızılkaya, O. (2025). Economic and geopolitical risk factors on environmental sustainability in Türkiye: a time-varying and wavelet coherence analysis. *Quality & Quantity*, 1-21. https://doi.org/10.1007/s11135-025-02442-x

Yılmaz, A. (2024). Do Geopolitical Risks and Political Stability Drive Foreign Direct Investments? New Evidence from Dynamic Panel CS-ARDL Model. *Ekonomi Politika ve Finans Araştırmaları Dergisi*, *9*(1), 61-87. https://doi.org/10.30784/epfad.1405599

Yu, M., & Wang, N. (2023). The influence of geopolitical risk on international direct investment and its countermeasures. *Sustainability*, *15*(3), 2522. https://doi.org/10.3390/su15032522

Zhang, B., Ma, J., Khan, M. A., Repnikova, V., Shidlovskaya, K., Barykin, S., & Ahmad, M. S. (2023). The effect of economic policy uncertainty on foreign direct investment in the era of global value chain: Evidence from the Asian countries. *Sustainability*, *15*(7), 6131. https://doi.org/10.3390/su15076131

Conflict of Interest: None.

Funding: None.

Ethical Approval: None.

Author Contributions: Faruk Mike (35%), İbrahim Ethem Akyıldız (35%), Oktay Kızılkaya (30%)

Çıkar Çatışması: Yoktur.

Finansal Destek: Yoktur.

Etik Onay: Yoktur.

Yazar Katkısı: Faruk Mike (%35), İbrahim Ethem Akyıldız (%35), Oktay Kızılkaya (%30)